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Alzheimer's Disease Detection Using Deep Learning

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***Abstract* - Alzheimer disease (AD) is a neuronal disease, which primarily affects memory, cognition and behavior. The early and correct diagnosis of the disease is highly important in the delaying of the stage and treatment of the patient. There is a strong potential of deep learning techniques (particularly the Convolutional Neural Networks (CNNs)) in medical image analysis, which has increased over the last several years. The analysis presents a CNN-based solution to the automatic identification and classification of the Alzheimer disease basing on the T1-weighted magnetic resonance imaging (MRI) scans. The provided model divides the brain MRI images into four clinical groups with Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. Image processing and data augmentation are carried out to enhance greater model robustness and generalization. In the course of the experiment, the suggested approach has been proven to be reliable in conducting a multi-class as well as a binary classification task with the help of bely. These findings suggest that the MRI analysis by means of deep learning can be offered as a supplement to the clinician in terms of an early diagnosis and staging of the Alzheimer disease.**

Keywords- Alzheimer disease, Convolutional Neural Networks, Deep learning, MRI, Medical image classification, Dementia.

I. Introduction

Alzheimer disease (AD) is an irreversible and chronic nervous disease, a result of which is the gradual loss of memory, sanity, language, and the pattern of behavior. As the disease advances, individuals with it become unable to manage their daily tasks and responsibilities alone, which reduces the quality of their life, and the reliance is then on other individuals. One of the most significant problems of the global population health is Alzheimer disease because of the high growth of the number of working-age individuals.

Early detection of the alzheimer disease is necessary because the therapeutic intervention would have a better chance of working at the initial stages of the cognitive deficiency. The most common traditional diagnostic activities entail cerebral exam, neurological and image survey. The Magnetic Resonance Imaging (MRI) has been the most widely utilized due to its ability to provide more structural information on the brain like the atrophic loss of the hippocampal and reduction of the cortex. Handi interpretation of MRI scans is however time consuming and highly dependent on the clinical wisdom which may yield diverse diagnosis and deliberation slowness.

Automated medical image analysis methods have been introduced by the recent advancements in the area of artificial intelligence (AI) and machine learning. The field of machine learning, specifically deep learning has drawn a significant amount of attention, due to the fact that it is able to learn complex representations of features, directly on raw data. Convolutional neural networks (CNNs) prove particularly beneficial because of the images, as they can explore the hierarchies and fine structures in them, which, under most circumstances, cannot be perceived by ordinary viewers.

The overall objective of the proposed research is to develop a CNN-based schema of the automated recognition and classification of Alzheimer disease using brain MRI scans.The general aim of the research would be to develop an efficient deep learning model capable of detecting and classifying MRI scans according to four clinically significant stages of dementia. The suggested system will be effective in assisting clinicians to diagnose the disease at an earlier stage, track and make superior clinical decisions based on the provision of stage-based classification.

II. Literature Review

The extraction of feature that was done by hand making use of classical algorithms in machine learning such as support vectors machine ability and k nearest neighbors have been the main additions to the old ways used to analyse medical images. These methods were reasonably good but were unable to work effectively due to the quality of the features designed manually and the inability to identify intricate non-linearities in medical images.

Deep learning came into the scene as a significant disruptor of computer vision and medical image processing. Convolutional Neural Networks (CNNs) are hierarchical partially feature representations of raw image data that are trained automatically, i.e. hand-free feature engineering is not needed. LeCun et al. demonstrated that CNNs are efficient in the image recognition process, which conditioned their massive application in medical imaging.

The CNN based models have also been applied successfully in healthcare where the models were used to determine the presence of these diseases in the lung, diabetic retinopathy, and the identification of brain tumor segments. By the virtue of these successes, scientists felt the need to explore the area of neurodegenerative diseases diagnosis using deep learning technologies. Regarding the Alzheimer disease, several studies have employed the use of the structural MRI data as a tool of differentiating healthy individuals and those with different stages of dementia.

Suk and colleagues proposed the use of an artificial intelligence approach that uses the multimodal imaging data to distinguish the Alzheimer disease and the mild impaired cognitive. Basaia et al. established that trained deep neural networks when using structural MRI scans were successful to differentiate between Alzheimer and cognitively normal individuals. Despite the promises, the small size of the dataset, the lack of balance between the classes, the differences across the scanners, and the number of small differences between the initial stages of the diseases have not yet been solved. Such challenges are why it is worth employing strong preprocessing techniques, tissue augmentation strategies, and reasonable CNNs.

III. Methodology

The suggested system is structured into the pipeline that includes data preparation, image preprocessing, design of the CNN model, training, and evaluation. Each of the stages is meant to be reliable, reproducible and perform effectively.

Figure 1 shows the best methodology for the proposed project model.

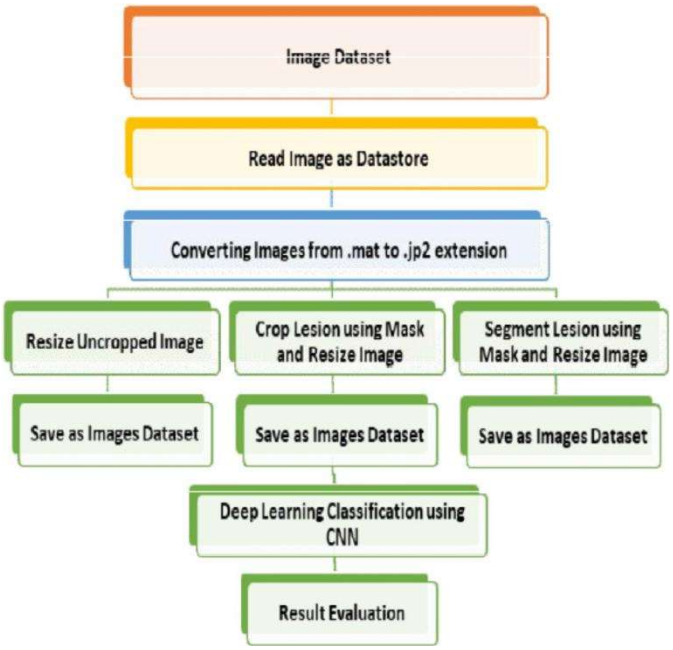


Fig.1. Proposed Methodology

A. Dataset Description

The data employed in this research was accessed in publicly accessible repositories of MRI of Alzheimer. It is a T1-weighted MRI of the brain that is divided into four categories, which include Non-Demented, Very Mild Demented, Mild Demented and Moderate Demented. In order to guarantee objective assessment the data were split into training, validation and testing sets.

Table 1: Summary of Used Image Data



B. Data Preprocessing

The preprocessing stage is an extremely crucial part of the medical images analysis as MRI scans tend to vary in size, resolution and intensity. The pictures were also resized to 224 x 224 to assure that the dimensions of the input would be standardized and therefore make the calculations less complex. The pixel values were scaled to create the range of 0-1 which improves the numerical stability of the training.

A data augmentation technique (rotation, horizontal) was used to augment the data so as to make the model more generalized and reduce the likelihood of overfitting.It was flipped, zoomed, and changed in terms of brightness.

Through these methods, the model is open to a variety of variations of the same data, so it is able to learn more robust feature representations. One-hot encoded vectors were used to accommodate multi-class classification because the labels of the classes were transformed into vectors.

C. CNN Architecture

The CNN model suggested is based on the EfficientNetB0 architecture and is performed by the Keras framework of deep learning. The network consists of several convolutional layers that extract features with max-pooling layers that that are used to reduce the spatial dimensions. To stabilize the training process and speed up its convergence, there were added batch normalization layers, and dropout layers were introduced to reduce overfitting.

- Conv2D Layers: 3 × 3 kernels for feature extraction with a ReLU activation.
- MaxPooling2D Layers: Dimensionality reduction layers to maintain only the most prominent features.
- Batch Normalization: Enhances model quality and speeds up convergence.
- Dropout Layers: Regularization in order to handle overfitting.
- Dense Layers: Fully connected layers for classification.
- Softmax Output Layer: Probabilities for four Alzheimer's categories.

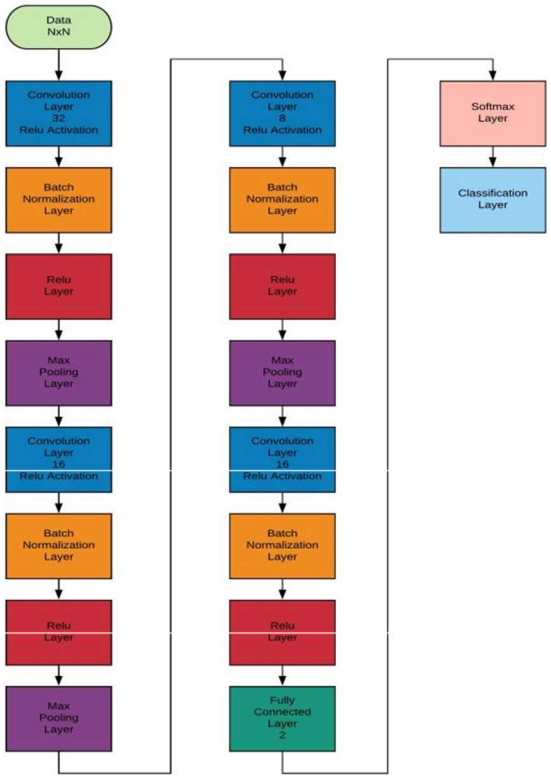


Fig.2. Proposed CNN Architecture

D. Training Parameters

Adam optimizer and categorical cross-entropy were used as the loss and learning rate respectively to train the model. The batch size was determined as 32 and 15 epochs were used to train the model. Early termination and model checkpoint call backs were used to avoid overfitting and maintain the best performing model.

- Optimizer: Adam optimizer(learning rate: 0.001)
- Loss Function: Cross - Entropy categorical
- Callbacks: Early stoping, and Model checkpoint for overfitting and to save the best model.
- Epochs: 15
- Batch size: 32

IV. Experimental Results

The effectiveness of the proposed CNN model was determined by conducting an independent test involving an independent test dataset to ascertain the performance of the model in classifying the stages of the Alzheimer disease. The use of standard evaluation metrics was done to check that it gives a holistic analysis of predictive ability of the model.

- *Precision:* The proposed model achieved a cumulative 83 percent of the precision in multi-classification on the four types of the Alzheimer disease, which were listed as Non-Demented, Very Mild Demented, Mild Demented and Moderate Demented. In addition, the accuracy also increased to 92 in case when the task was narrowed down to binary classification (Demented vs. Non-Demented) and this suggests that the model has a high possibility to distinguish between the healthy and the affected persons.

- *Precision, c Recall and F1-Score:* In order to test the reliability of the classification, as well, precision, recall and F1-score of each of the classes were also retrieved. The obtained results suggest that the precision, as well as the recall values were at all times greater than the 90% mark and it can be concluded that the model is efficient and can minimize the false positives as well as the false negatives. F1-score was also over 90 as a compromise between precision and recall and the results were balanced and consistent across the entire stages of the disease.

- *Confusion Matrix Analysis:* The confusion analysis was performed to analyse the behaviour of the prediction in terms of classes. The results indicate that the majority of the forecasts fell on the correct side. The greatest instances of misclassification of minor cases were found between the Mild Demented and the Very Mild Demented. This comes not as a surprise, as the changes that take place in the brain during these early stages are so slight and difficult to detect. All in all, the strength of the proposed model is demonstrated by the confusion matrix.

A. *Training and validation performance*

The training and validation performance should be evaluated by comparing the original performance with the obtained performance with the training (Bartl et al., 2005).<|human|>D. Training and Validation Performance: The training and validation performance must be assessed by comparing the initial performance with the performance obtained through the training (Bartl et al., 2005).

```
83s 4s/step - accuracy: 0.5971 - loss: 1.0376 - val_accuracy: 0.5934 - val_loss: 0.8793
5 122ms/step - accuracy: 0.7437 - loss: 0.5869 - val_accuracy: 0.6411 - val_loss: 0.7640
5 121ms/step - accuracy: 0.7833 - loss: 0.5070 - val_accuracy: 0.6794 - val_loss: 0.7075
5 122ms/step - accuracy: 0.7952 - loss: 0.4716 - val_accuracy: 0.6888 - val_loss: 0.6803
5 124ms/step - accuracy: 0.8145 - loss: 0.4277 - val_accuracy: 0.7138 - val_loss: 0.6302
5 121ms/step - accuracy: 0.8214 - loss: 0.4169 - val_accuracy: 0.7287 - val_loss: 0.6142
5 122ms/step - accuracy: 0.8390 - loss: 0.3785 - val_accuracy: 0.7608 - val_loss: 0.5866
5 122ms/step - accuracy: 0.8442 - loss: 0.3724 - val_accuracy: 0.7670 - val_loss: 0.5626
5 122ms/step - accuracy: 0.8525 - loss: 0.3555 - val_accuracy: 0.7866 - val_loss: 0.5325
5 121ms/step - accuracy: 0.8582 - loss: 0.3470 - val_accuracy: 0.7944 - val_loss: 0.4970
5 122ms/step - accuracy: 0.8664 - loss: 0.3290 - val_accuracy: 0.8092 - val_loss: 0.4842
5 121ms/step - accuracy: 0.8677 - loss: 0.3180 - val_accuracy: 0.8022 - val_loss: 0.4843
5 122ms/step - accuracy: 0.8815 - loss: 0.2942 - val_accuracy: 0.8249 - val_loss: 0.4407
5 124ms/step - accuracy: 0.8784 - loss: 0.2938 - val_accuracy: 0.8194 - val_loss: 0.4506
5 122ms/step - accuracy: 0.8816 - loss: 0.2867 - val_accuracy: 0.8335 - val_loss: 0.4229
```

Fig.3. Accuracy and Loss through epochs

The training and validation accuracy curves demonstrate that the successive epochs exhibit a progressive improvement in the accuracy values which is a sign of successful learning behavior. There was an improvement in training accuracy with validation following the same trend with a slight difference indicating positive generalization. On the same note, training and validation loss values reduced smoothly, with a stable convergence with no serious overfitting.

B. *Visualization*

The visualization of the performance was done by accuracy and loss curves of the training and validation phases. The plots depict that the model greatly converged within the desired number of epochs. Moreover, sample test MRI images have been tested with the trained model and the results that are obtained were mostly similar to the ground truth labels, which further confirm the usefulness of the model in practice.

Accuracy curve graph and Loss curve graph are shown in Figure 4 and Figure 5 respectively.

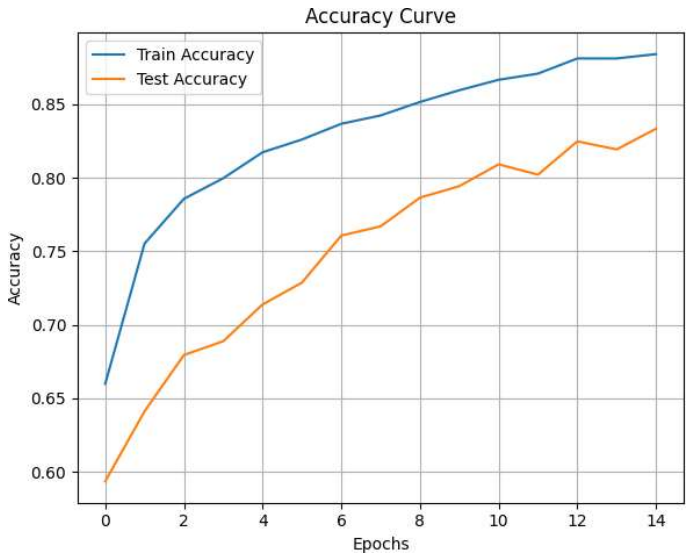


Fig.4. Curve graph of accuracy

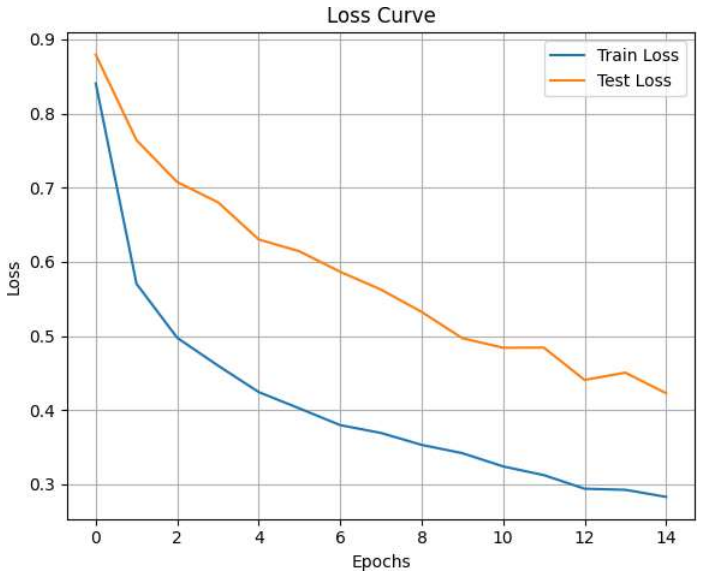


Fig.5. Curve graph of loss

V. Discussion

The research results using the suggested CNN-based model prove that it is possible to successfully implement the deep learning to detect and diagnose the Alzheimer disease using the MRI images of the brain. The model had a high level of performance in binary and multi-class classification, which indicates that the model can classify a healthy individual and patients with different stages of dementia. These findings are in line with the prior research that advises usefulness of CNNs in the sphere of medical imageries and in the diagnosis of the Alzheimer disease (Suk et al., 2014; Basaia et al., 2019).

The overall findings of the different stages of the dementia suggest that the model is very eloquent in attaining structural systems patterns of the brain that relate to the illness course. However, it was a bit more difficult to place the categorization between the first stages that is the Very Mild Demented and the Mild Demented. It is not an unexpected problem because structural changes in the brain during the early phases of the progression of Alzheimer are typically irrelevant and difficult to detect even in clinical experts (Pereira et al., 2016).

A. Strengths of the Model

- *Very Good Presentation of the content:* The proposed model in question has achieved an accuracy of 83 and 92 in multi-class and binary classification respectively. The findings such as these are indicative of the fact that the model can be successfully employed to find out the presence of the Alzheimer disease as well as the rates of its severity. It is also very accurate, and this is a reflection of how the model can identify meaningful objects of MRI images as it has been the case in the previous CNN-based research (LeCun et al., 1998; Shen et al., 2017).

- *Learning:* The individual is taught through generalization. The slow increase in the accuracy of the training and a moderate decrease in training and validation loss has demonstrated that the CNN has learnt with success on the data. This shows that there is massive generalization into invisible MRI images rather than memorizing of training set. Overfitting was also minimized and regularization increased model stability in addition to being true in other past studies done on deep learning in medical imaging (Basaia et al., 2019).

B. Threats and Weaknesses of model.

- *Hardly Makes it through Classification at the Brief Level:* The analysis of the confusion matrix showed that the migration of the Very Mild Demented and the Mild Demented groups was slight. The factors that have brought about this dilemma include the fact that structural differences between these early stages of the Alzheimer disease are limited and they cannot be filtered. The present literature also outlines the same limitations because the detection of Alzheimer in its early stages is a complex procedure that is grounded solely on MRI data (Suk et al., 2014).

- *Variability in MRI Data:* The variation in the MRI images between scanners and institutions in terms of resolution, contrast and acquisition setting can be very large. Such difference can be disruptive to the integrity of deep learning models. Although this study is already on the way to using the principles of normalization and preprocessing, the further use of MRI data acquired in a more diverse pool of sources would increase the quality of reliability and generalization of the proposed model (Pereira et al., 2016).

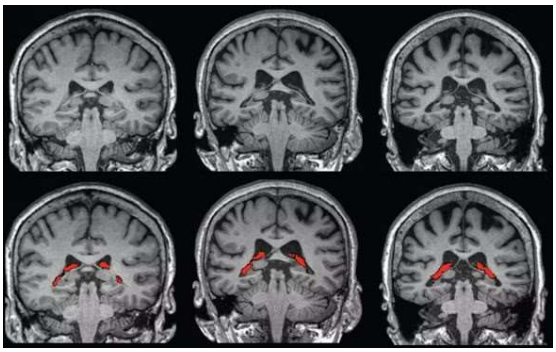


Fig.6. Variation in MRI Dataset

C. Future Directions

- *Model Enhancement:* One can dedicate future research to the more complicated CNN models or other transfer learning methods, such as VGG16 or ResNet50, to improve the accuracy of the feature analysis and classification. The models have demonstrated positive efforts in various medical imaging and can be exploited to improve the detection of Alzheimer at its early stages.

- *Data set and Multimodal Data: Expansion:* It would also provide more diagnostic data in the introduction of a larger dataset and including multimodal imaging data such as PET or functional MRI (fMMRI). Combination of various imaging has shown better classification and power in the case of the Alzheimer disease studies (Suk et al., 2014).

- *Clinical Validation:* The proposed model is supposed to be used in the clinical setting and is intended to be tested on the real-life patient statistics. The practicality, reliability, and usefulness of the system as a useful diagnostic tool in the healthcare setup will be determined by the clinical validation.

VI. Conclusion

These are some of the ethical and comprehensive deep learning-based frameworks of the detection and classification of Alzheimer disease in structural MRI images as presented in this paper. Using a CNN architecture and powerful methods of preprocessing, the suggested model has a high performance when it comes to the detection of various stages of the dementia condition.

The results indicate that the deep learning-based MRI analysis could be used as an effective decision-support tool by clinicians to help in the early diagnosis and disease monitoring. The future directions of work will be to involve more sophisticated deep learning structures, increase the diversity of data, and test the suggested system in real clinical conditions.

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